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# Observations of resonant modes formation in microwave generated magnetized plasmas

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**Abstract.** Ion sources have a significant number of applications in accelerator facilities and in industrial applications. In particular, the electron cyclotron resonance ion sources (ECRIS) are nowadays the most effective devices that can feed particle accelerators in a continuous and reliable way, providing high current beams of low and medium charge state ions and lower, but still remarkable, beam current for highly charged ions. In recent years several experiments have shown that the current, the charge states and even the beam shape change by slightly varying the microwave frequency (the so-called frequency tuning effect – FTE). The theoretical explanation of these results is based on the difference in the electromagnetic field pattern over the resonance surface, i.e. that region where the electrons resonantly interact with the incoming wave. In order to be consistent with the experiments, this model requires that standing waves are formed also in presence of a dense plasma. The proof was sought by means of a series of measurements performed with a network analyzer and with a plasma reactor operating at 2.45 GHz, according to the principles of the microwave discharge ion sources (MDIS). The measurements have been carried out with the aim to achieve the electromagnetic characterization of the plasma chamber in terms of possible excited resonant modes with and without plasma, and they reported that resonant modes are excited inside the cavity even in presence of a dense plasma. It was observed that the plasma dynamics strongly depends on the structure of the standing waves that are generated. The measurement of the eigen-frequencies' shifts were carried out for several values of pressure and RF power, thus linking the shift with the plasma density measured by a Langmuir probe. The changes in plasma shape, density and electron temperature have been also monitored for different operating conditions. A strong variation of plasma properties has been observed as a consequence of the introduction of the Langmuir probe inside the resonant cavity, thus demonstrating that the standing wave can be strongly perturbed even by means of relatively small metallic electrodes. The measurements reported hereinafter are relevant also for ECRIS, because they confirm the validity of the theoretical model that describes the frequency tuning.

## 1 Importance of frequency tuning effect in ion sources: theoretical and experimental evidences

Many experiments in the last years have shown that significant improvements of ECRIS performances, both in terms of the extracted currents and charge states, are obtainable by choosing the appropriate microwave frequency. In particular, recently, it has been demonstrated that changes in the ECRIS performances (in terms of current produced and beam shape) are obtained by slightly varying the microwave feed frequency. This phenomenon is usually defined as the 'frequency tuning' effect (FTE) and it strongly affects the beam formation dynamics [1]. The effect of the microwave frequency scaling on the plasma properties (and then on the beam current and mean charge state) is well known from the scaling laws [2]. These rules are based on semi-empirical considerations about the plasma density scaling with the frequency. Something new occurs in case of the FTE, as it produces strong fluctuations of the beam performances even in MHz ranges of frequency variation. For example, in the experiment reported in [1] a considerable increase of the extracted current was observed even for frequency variations in the range of 50 MHz. Theoretically the simulations have demonstrated that the ECR heating efficiency depends on the electromagnetic field distribution and on its value over the resonance surface: a completely different electromagnetic field pattern is produced if the frequency is slightly changed [3]. However, clear experimental evidences of standing wave formation in non-uniform and non-isotropic plasmas have not been available up to now. Preliminary investigations have been carried out at INFN-LNS for the SERSE superconducting ion source [4]. Several measurements were performed, with and without plasma, to characterize the system (plasma chamber plus microwaves line) in terms of the S matrix.

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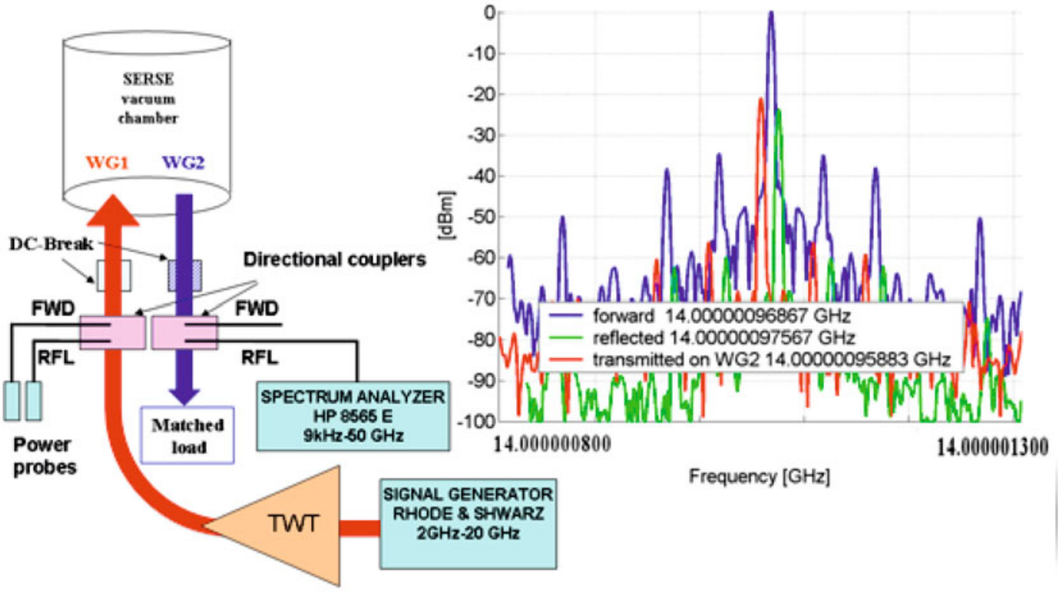



**Fig. 1.** (Color online) The experimental setup (right) and the spectra (left) of the forward and reflected power on WG1 and WG2.

The analysis of the data without plasma have clearly shown the typical behaviour of a resonating structure [5] and a frequency dependent structure was observed even in presence of plasma. These results triggered a theoretical investigation, based on the classical theory of cavity resonators; the microwave coupling to the SERSE plasma cavity and the modes excitation within it were considered along with their dependence on the source waveguide inputs [6]. The analysis of the ECR heating process is the basis of the experimental activity described hereinafter.

## 2 Preliminary observations in ECRIS

The campaign of measurements performed in 2005 in order to characterize the SERSE source from the electromagnetic point of view, gave a first clue about the standing wave formation in ECRIS. The setup is shown in Figure 1: the SERSE source was coupled to a TWT through a WR62 waveguide containing a dc-break and a directional coupler, while the other waveguide was terminated on a matched load just after the directional coupler. Two power probes connected to the directional coupler on the WG1 waveguide permitted us to reconstruct the reflection coefficient over a wide range of frequencies. A spectrum analyzer was also employed to determine the spectra of the emitted and reflected waves. The measurements without the plasma [5] have shown the typical frequency response of a highly resonating structure, however, due to the poor directivity of the couplers we were not able to correctly reconstruct $S_{11}$ and $S_{22}$ with the plasma. Nevertheless, the observations with the spectrum analyzer (exploiting the peculiarity of SERSE to have two separate waveguide inputs) gave an evident clue of standing wave formation in the plasma chamber. In particular we were able to reveal on the WG2 directional coupler (located almost 3 m behind the source injection flange) a spectrum identical to the ones of the forward and reflected signal on WG1. As we are using a waveguide to pick up the signal from the chamber (and not a coaxial probe), it means that a mode travelling on the $z$ axis is present in the plasma chamber of SERSE and that is coupled to WG2; it gives a clear hint of standing wave formation inside the plasma chamber. Therefore, in order to undertake further investigations on such subjects we decided to employ a plasma reactor used at INFN-LNS for environmental purposes.

## 3 Observations in microwave discharge type ion sources

The plasma reactor was designed and built at INFN-LNS (Figs. 2 and 3) as a tool to study the dissociation of complex molecules. It is a microwave discharge plasma source [7,8], with a plasma that is weakly ionized and strongly collisional because of low electron temperatures ($T_e \sim 1{-}10$ eV) and high pressures (0.1−1 mbar). The plasma chamber is a stainless steel cylinder 268.2 mm long, with a radius of 68.5 mm; a NdFeB permanent magnet generates an off-resonance magnetic field along the plasma chamber axis (with a maximum of 0.1 T on axis). The high electron-neutral collision frequency determines a high rate of molecules dissociation. For safety reasons, the reactor efficiency has been tested by using non dangerous gas, in particular n-hexane ($C_6H_{14}$) and cyclehexane ($C_6H_{12}$) [8] but the final aim of the experiment is the dissociation of dangerous complex molecules (e.g. dioxine and PCB). Several measurements have been carried out in order to determine the optimum operating conditions in terms of microwave power and gas pressure and in 2007 n-hexane and cyclo-hexane have been successfully dissociated [8]. In the injection side of the reactor the microwave

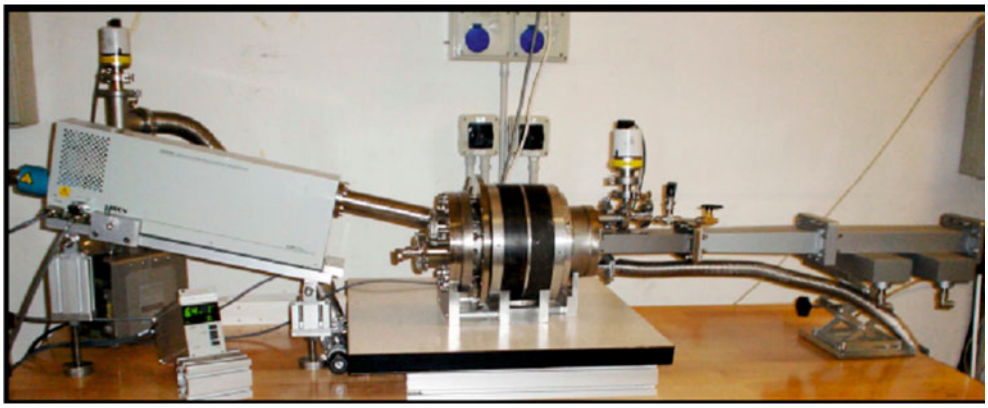


**Fig. 2.** (Color online) The plasma reactor on its test-bench at INFN-LNS.

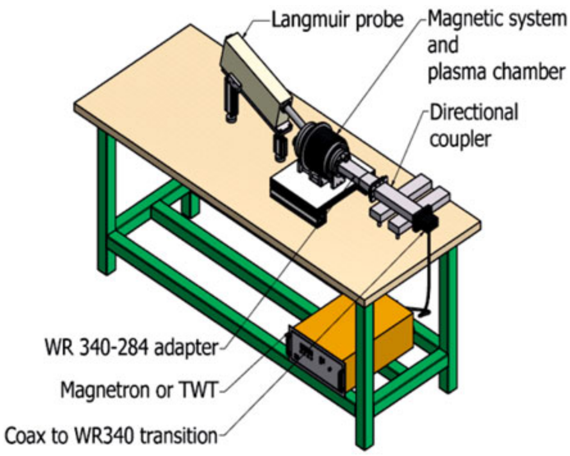



**Fig. 3.** (Color online) A scheme of the experimental setup.

injection aperture, the flanges for the pumping and for the gas injection are located. In the opposite side other flanges are available to connect the mass spectrometer, the optical window (to be used for plasma observation and for optical spectroscopy), a coaxial probe to pick up the signal from the chamber and the Langmuir probe.

The complete layout of the experimental apparatus is shown in Figure 3: a magnetron working at 2.45 GHz with maximum power of 300 W is coupled to the plasma reactor through a WR284 waveguide line working in the $TE_{10}$ dominant mode. In order to validate the results obtained the variation of the plasma properties were checked in the range 2−5 GHz by using a broadband generator (TWT) in place of the magnetron. A rotative pump is used for vacuum (pressure down to $10^{-2}$ mbar are usually obtained). This setup was chosen as a testbench to investigate the standing wave formation inside the resonant cavity in presence of plasma. A network analyzer connected to the coaxial connector on the injection side of the reactor permitted to determine the S matrix with and without plasma and to investigate the best condition of wave-plasma coupling. In our experiment we fixed the microwave feeding the plasma at 2.45 GHz and analysed the plasma properties in the 1−3 GHz frequency range. For the $S_{11}$ parameter characterization the plasma was ignited through the waveguide and the network analyzer was connected to the coaxial connector. By analyzing the $S_{11}$ parameter it is possible to characterize the modes that exist inside the plasma chamber, as they correspond to frequencies for which the minimum values of the $S_{11}$ occur. Measurements of scattering parameters have been carried out in parallel with Langmuir probe measurements of electron temperature and density. Main features of the Langmuir probe diagnostics are discussed in [9]. The Langmuir probe used for our experiment consists of a 4 mm length of tungsten wire, with a diameter of 150 $\mu$m (the metallic wire protrudes from a cylindrical alumina sheath). A motor allows to move the probe inside the plasma chamber with an accuracy of 0.1 mm. The Langmuir probe is located so that it can penetrate inside the cavity for its entire length, forming an angle of 14° with respect to the chamber axis, as shown in Figure 2. A commercial Langmuir probe has been used [10], that allows to easily determine the main plasma parameters by fitting the characteristic I-V curve. The particle collection by means of biased electrodes may be influenced by collisions and/or magnetic fields. Collisions work as a friction, thus limiting the detected current; eventual magnetic field effects may cause a non saturation of the ion current, even for extremely negative potentials applied to the probe tip. In order to verify the correct employment of the unperturbed theory we compared the electron gyroradius with the probe radius, assuming $B = 0.1$ T and an electron energy of 10 eV (which are typical values in microwave discharge plasmas). According to the relationship between the electron Larmor radius ($r_{Le}$), ion Larmor radius ($r_{Li}$) and the tip radius ($r_p$) we can have three regimes:

$$\begin{cases} r_p \ll r_{Le} & \text{weak } B \text{ field regime} \\ r_{Le} < r_p < r_{Li} & \text{moderate } B \text{ field regime} \\ r_{Li} < r_p & \text{strong } B \text{ field regime.} \end{cases}$$

In our case $r_{Le} \simeq 1.2 \times 10^{-4}$ m that is larger than the probe radius $(r_p) = 7.51 \times 2 \times 10^{-5}$ m. Hence the first condition is satisfied and we can consider the particle collection unperturbed by the magnetic field. About the collisions, the unperturbed theory is valid as long as the mean free path is larger than probe radius and Debye length.

In our case, assuming a plasma density of $10^{10}$ cm$^{-3}$ and the above mentioned temperature,the mean free path results three times the Debye length, thus the conditions on collisionless particle collection are fulfilled.

## 4 Standing waves in plasmas: the state of art

The determination of the resonant frequency shift and the measurement of the quality factor Q of the cavity allow to calculate the plasma density and the collision frequency [11,12]. In particular Adgur has given a complete picture of the mode pattern existing in a plasma filled cylindrical cavity, including modes that are not directly related to the ones existing in an empty cavity. However, he considers a central plasma column, whereas the device described in Section 3 is somewhat different. In fact the plasma completely fills the cavity and the magnetic field makes it a non-isotropic medium for the electromagnetic field propagation [9]. Additional hints come from Asmussen [13], who studied the interaction of microwaves and cavity in a plasma source, and he observed that the plasma density is not proportional to the incident power. Only a small fraction of the additional incident power is absorbed into the plasma, the remaining part reflected from the cavity. This is a fundamental problem when attempting to produce a high-density plasma inside a microwave cavity and it can be solved by varying the excitation frequency while the plasma is sustained by radiofrequency, or by varying the cavity dimensions. The Asmussen's model explains the saturation of electron density with the pressure. A device more similar to ours was investigated by Shohet et al. [14], as he assumed a multiple-shell model considering a cylindrical plasma column of variable radial density filling the entire plasma chamber. As mentioned before, the main difference between our plasma reactor and the other plasma sources above described is connected with the presence of the magnetic field. It is not possible to determine a priori whether the presence of the electron cyclotron resonance destroys the mode structure. The electron heating in ECR devices is a stochastic phenomenon [15]: at the beginning the electrons are simply accelerated by the rotating electric field associated to the pumping microwave. During the initial stage, the electron motion in the wave field is adiabatic. After few ns non adiabatic effects arise, and the electron acceleration changes into heating, i.e. the impulse randomization, that is normally provided in thermodynamic systems by collisions, is here governed by these non-adiabatic effects resulting from the wave-particle interaction. A wave travelling in a given direction inside the cavity exchanges stochastically energy with plasma electrons. This non-collisional dissipative energy absorption mechanism permits the wave to reach the opposite flange of the plasma chamber, thus being reflected backward. It is just the reflection that is at the basis of possible standing wave formation. This speculative discussion is confirmed by the preliminary experiments carried out on SERSE source previously reported. Microwave discharge plasma sources are quite different devices with respect to the ECRIS, as for the plasma energy content, but the plasma densities are quite similar. Experimental measurements of the electron density and temperature on the device described in Section 3 have put in evidence that they have a radial and axial variation: the plasma density is usually higher in proximity of the waveguide feeding the cavity [9], and the plasma radius is comparable with the chamber's one. It is not easy to link the electron density and the mode shift, but in the following, we have made some attempts to determine theoretically the shift of resonance frequencies.

## 5 Electromagnetic waves propagation in a resonating cavity

A theoretical calculation of the resonant modes in vacuum has been performed in parallel with the experiment, by solving the eigenvalue equations for the electromagnetic field inside the plasma chamber and by using an oversimplified model of the cavity (it has been considered empty and completely closed). In a cavity with radius $a$ and length $l$, filled with a medium having dielectric constant $\epsilon_r$ and relative permittivity $\mu_r$, the equations that determine the allowed frequencies for TM and TE modes are [16]:

$$f_{n\nu r} = \frac{c}{2\pi\sqrt{\epsilon_r\mu_r}}\sqrt{\left(\frac{x_{n\nu}}{a}\right)^2 + \left(\frac{r\pi}{l}\right)^2} \tag{1}$$

$$f_{n\nu r} = \frac{c}{2\pi\sqrt{\epsilon_r\mu_r}}\sqrt{\left(\frac{x'_{n\nu}}{a}\right)^2 + \left(\frac{r\pi}{l}\right)^2} \tag{2}$$

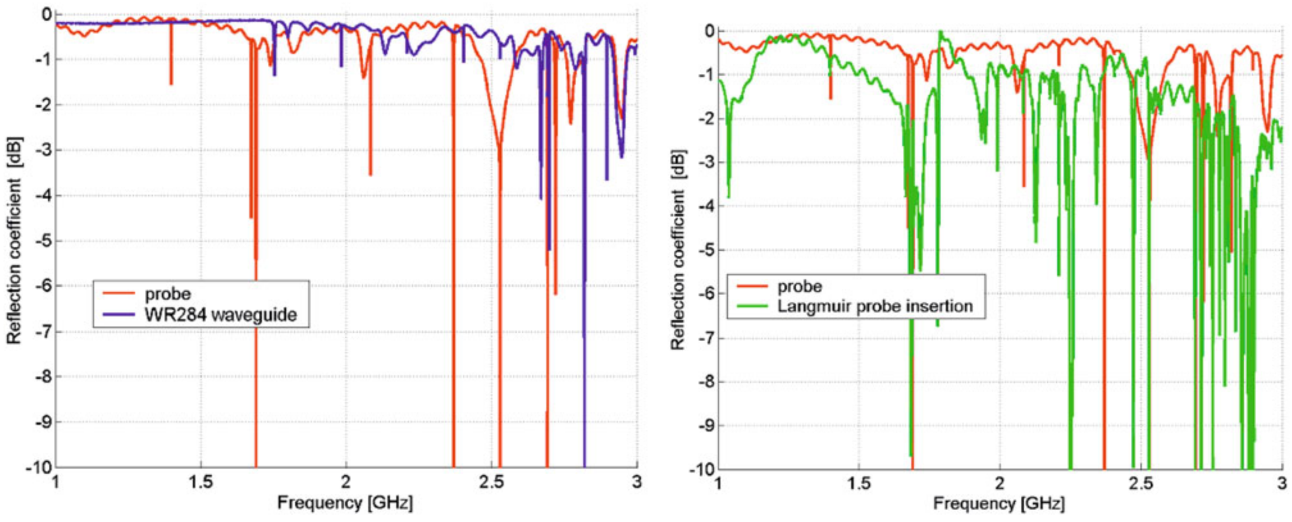
where: $x_{n\nu}$ and $x'_{n\nu}$ are respectively the zeros of order $\nu$ of the Bessel functions of order $n$ and its first derivative. Then the three indices $n$, $\nu$, $r$ identify the electromagnetic field pattern of each mode; $l$ is the plasma reactor chamber length (268.2 mm) and $a$ is the radius (68.5 mm). By supposing the chamber is filled with a homogeneous and un-magnetized medium the corresponding electrical permittivity is given by the relation:

$$\epsilon_r = 1 - \left(\frac{\omega_p}{\omega}\right)^2 \tag{3}$$

where $\omega$ is the pulsation generating the plasma (2.45 GHz) and $\omega_p$ is the plasma pulsation:

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \epsilon_o}} \tag{4}$$

where $m_e$ and $e$ are respectively the electron mass and electron charge, $\epsilon_o$ is the electrical permittivity in vacuum and $n_e$ is the electron density. Therefore, under these hypotheses, the plasma buildup can be seen as a change in the electrical permittivity and therefore as a change in the resonant frequencies, because of the change of $\epsilon$ in equations (1) and (2). A frequency shift has been observed in the plasma reactor as described in the following section.




(a) (b)

**Fig. 4.** (Color online) (a) $S_{11}$ parameter obtained in vacuum conditions through the microwave probe (red line) and the WR284 waveguide (blue line). (b) Comparison between the $S_{11}$ parameter measured in vacuum conditions with (green line) and without (red line) Langmuir probe inserted into the chamber.

## 6 Standing waves detection in the plasma chamber

A detailed characterization of the scattering parameter $S_{11}$ has been carried out in vacuum at the beginning (Fig. 4a), then in the presence of plasma generated at different power and pressure. The penetration of the Langmuir probe inside the plasma reactor chamber significantly affects the reflection coefficient measured at the coaxial connector as shown in Figure 4b. The probe has been positioned for the plasma parameters measurements at $z = 17.2$ cm inside the cavity and it consists in a 7 mm diameter alumina cylinder penetrating inside the cavity with a slope of 14° with respect to the plasma chamber axis.

The measured mode depends on the frequency of the electromagnetic wave launched in the plasma chamber and on the matching between the plasma chamber and the waveguide. Table 1 shows a comparison between the calculated resonant frequencies and the measured ones by the network analyzer connected to the microwave probe and to the WR284 waveguide.

Note that many of the calculated modes are excited and only minor differences exist with the measured frequencies due mainly to the presence of many 'holes', i.e. the flanges, which make the plasma reactor chamber a non-ideal resonant cavity. After this characterization in vacuum conditions we created the plasma through the waveguide using a 2.45 GHz magnetron and measured $S_{11}$ by connecting the network analyzer to the microwave probe. Since from the first the measurements it was clear that the plasma creation does not destroy all the resonant modes and that a resonating structure is present even with plasma inside the reactor. Figure 5 features the differences in the $S_{11}$ parameter with and without plasma for a microwave power of 70 W and a pressure of 0.7 mbar. Some modes are shifted in frequency by the plasma, some others disappear and a few modes appear only in presence of the plasma. Therefore, as a consequence of plasma formation, the $S_{11}$ structure strongly changes as a reult of coupling changes between magnetron and reactor, however some clear resonances are maintained.

**Table 1.** Comparison between the calculated modes and the measured ones.

| | Mode | Calculated resonance frequency (GHz) | Measured resonance frequency probe (GHz) | Measured resonance frequency WR284 (GHz) |
|---|---|---|---|---|
| 1 | $TE_{111}$ | 1.39896 | 1.3978 | – |
| 2 | $TM_{010}$ | 1.67507 | 1.6731 | – |
| 3 | $TE_{112}$ | 1.70123 | 1.6897 | – |
| 4 | $TM_{011}$ | 1.76585 | – | 1.7534 |
| 5 | $TM_{012}$ | 2.01378 | – | 1.9822 |
| 6 | $TE_{113}$ | 2.11092 | 2.0838 | – |
| 7 | $TE_{211}$ | 2.19960 | 2.2081 | 2.2089 |
| 8 | $TM_{013}$ | 2.37005 | 2.3687 | – |
| 9 | $TE_{212}$ | 2.40320 | – | 2.4033 |
| 10 | $TE_{114}$ | 2.57732 | 2.5288 | 2.5287 |
| 11 | $TM_{110}$ | 2.66896 | 2.6925 | 2.6925 |
| 12 | $TE_{213}$ | 2.70972 | 2.6975 | 2.6975 |
| 13 | $TE_{011}$ | 2.72685 | 2.7198 | – |
| 14 | $TM_{111}$ | As previous | As previous | As previous |
| 15 | $TM_{014}$ | 2.79351 | 2.8191 | 2.8191 |
| 16 | $TE_{012}$ | 2.89358 | 2.8956 | 2.8956 |
| 17 | $TM_{112}$ | As previous | As previous | As previous |

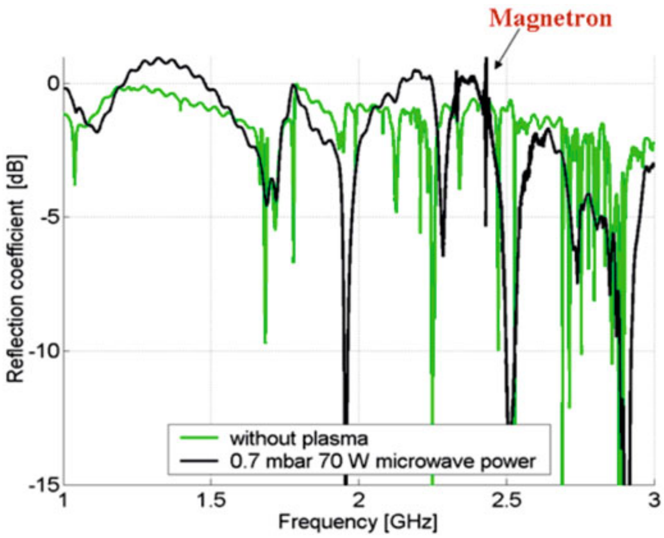



**Fig. 5.** (Color online) $S_{11}$ parameter with (black line) and without (green line) plasma.

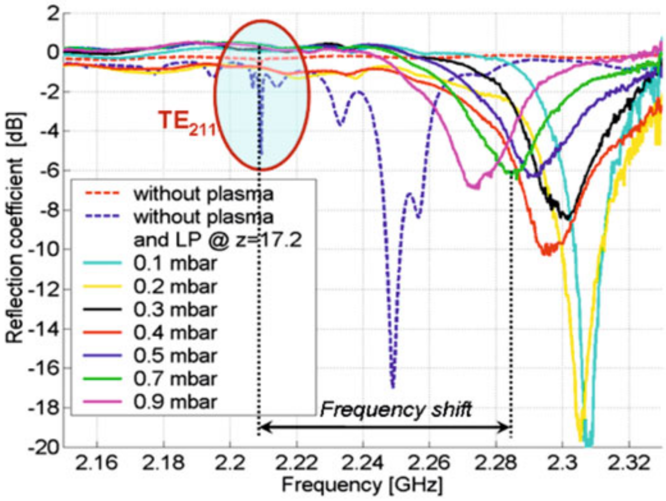



**Fig. 6.** (Color online) Shift of the resonant mode at 2.21 GHz for different values of pressures (power 70 W).

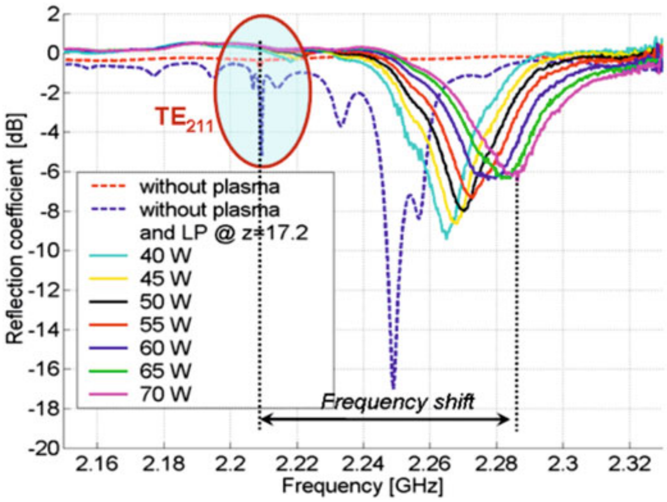



**Fig. 7.** (Color online) Shift of the resonant mode at 2.21 GHz for different values of power (pressure = 0.7 mbar).

The shift of the resonant modes with the RF power and pressure has been investigated and correlated with the electron density measured by means of a Langmuir probe. The LP insertion in the plasma chamber introduces some perturbations as shown in Figure 4b which makes unclear the reading of the frequency spectrum and therefore the identification of the mode shift when plasma is ignited. However, such identification can be achieved by means of the electron density measurement of the LP. In particular for 70 W of microwave power and 0.7 mbar of gas pressure an electron density value of $6 \times 10^{15}$ $\mathrm{m}^{-3}$ was measured with the probe at $z = 17.2$ cm. By inserting this value into equations (3) and (4) it is possible to reconstruct the mode frequency in vacuum conditions for the modes (see Fig. 5) at 2.28 GHz and 2.913 GHz. In particular for the first mode the frequency in vacuum is around 2.21 GHz. Measurements demonstrate that a mode is present before the LP insertion in correspondence with this frequency (see Tab. 1). By using the same value of permittivity for the mode at 2.913 GHz we determine that its vacuum frequency should be around 2.8 GHz. For our calculations we assumed that the shifting mode is the closer one with frequency of 2.8191 GHz. According to the theoretical determination of the resonant frequencies, we can link the above frequencies to two particular modes: the $TE_{211}$ and the $TM_{014}$ as it is reported in Table 1. As explained in [11], in case of anisotropic systems the resonant modes assume a hybrid nature, i.e. they cannot be strictly defined as TE or TM modes, and in our case this effect may occur especially when the probe is inserted into the plasma chamber. We are unable to make any conclusion about the mode at 2.5 GHz (see Fig. 5): the behaviour is equal to the one of the previous modes, but it is significantly affected from the spurious of the magnetron which operates at 2.45 GHz (the carrier is present in Fig. 5 as a narrow peak) and therefore it is not possible to quantify the frequency shift. The frequency shift was observed for different values of power and pressure. The microwave power range used for the experiment was 40−70 W and the gas pressure was vared between 0.1 and 0.9 mbar. The results are shown in Figures 6 and 7 for the $TE_{211}$ mode; similar trends were observed for the mode $TM_{014}$ near 2.81 GHz. The wave coupling changes as a consequence of the RF power and chamber pressure variations, which affect the electron density. The coupling changes are put in evidence by the variation of the peak height, which can change up to 10 dB by varying the power or the pressure.

A normalized shift has been determined by means of the following equation:

$$\frac{\Delta f}{f} = \frac{f_{on} - f_{off}}{f_{off}} \times 100 \quad (5)$$

where $f_{on}$ and $f_{off}$ are the frequencies with and without plasma. From Figures 8 and 9 it can be observed that the shift decreases for higher values of pressure and

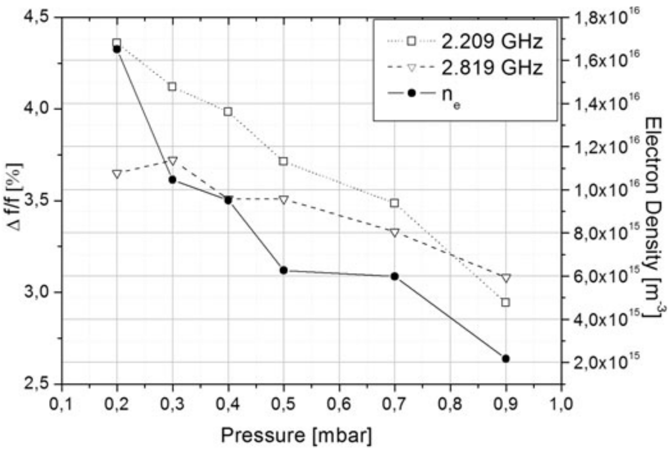



**Fig. 8.** Comparison between the frequency shift and the electron density trends for different background pressures at PRF = 70 W.

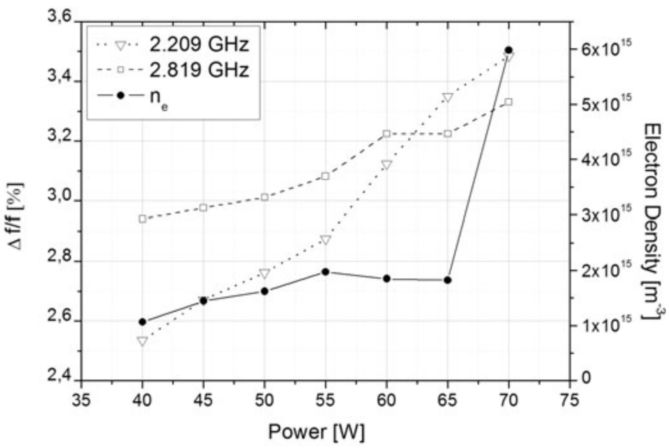



**Fig. 9.** Comparison between the frequency shift and the electron density trends for different microwave power at 0.7 mbar of background pressure.

it is greater for higher values of power. Such a trend is due to the variation of the electron density, which is well known from previous measurements to increase for higher power and to decrease for higher pressures [9]. Hence the figures clearly correlate the electron density and the frequency shift, which increases with the density, as expected from calculations in the case of homogeneous and isotropic plasma filling cylindrical cavities. During the interaction with the plasma the wave can be weakly or strongly absorbed or reflected at the plasma boundary if the pressure and/or the power are changed. For example, in some cases the formation of an upstream overdense plasma has been observed [9] due to a strong energy absorption at the first cyclotron harmonic. The electromagnetic field structure inside the plasma chamber is strongly biased by the electron density distribution too. The measurements of the electron density, in a fixed position of the plasma chamber, give an incomplete characterization of the plasma properties. Conversely, it is not possible to perform measurements in different regions of the cavity, as the variations of the probe position lead to remarkable variation in the

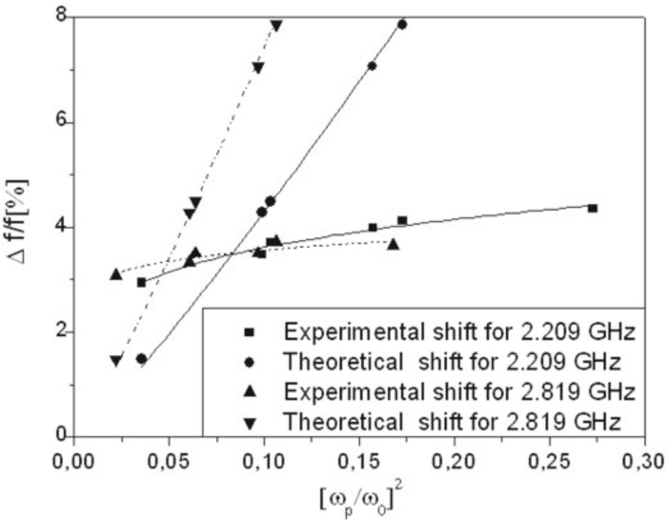



**Fig. 10.** Theoretical and experimental trends of the frequency shift with respect to the electron density for the two modes under investigation.

structure of the $S_{11}$ parameter. Then we take into account only partial measurements carried out in a position where the density is less affected by phenomena leading to strong changes of the plasma properties.

The trend of the frequency shift for the two modes with respect to the dimensionless plasma density $\eta$:

$$\eta = \left(\frac{\omega_p}{\omega}\right)^2 = \frac{n_e e^2}{m\epsilon_o \omega^2} \tag{6}$$

is shown in Figure 10. The theoretical shift calculated for isotropic and homogeneous plasmas, according to equations (1) and (2), is also shown in Figure 10.

The two experimental curves shown in Figure 10 have been fitted with functions of type:

$$y = ax^b. \tag{7}$$

For the mode near 2.209 GHz the $a$ and $b$ parameters of the fit are approximately equal to 5.7 and 0.2, while for the other mode they are 4.4 and 0.09 respectively. The formula (5) features that for both the modes no saturation occurs: this is in agreement with the modelling of completely filled cavities in the case of inhomogeneous plasmas, as described in [14]. The increase of the experimental frequency shift with the density is less marked with respect to the theoretical one, probably because of the strong axial density inhomogeneity, which leads to low density regions especially far enough from the microwave window. The magnetic field also plays a role, and the difference between the theoretical calculations and the experimental trend of the frequency shift can be considered as a sort of weight of the magnetic field and axial density variation induced effects on the electromagnetic field propagation inside the plasma reactor's cavity. A more detailed analysis can be performed on the basis of the frequency shift. We may suppose that the resonance cavity is filled with a homogeneous and unmagnetized plasma, then by means of

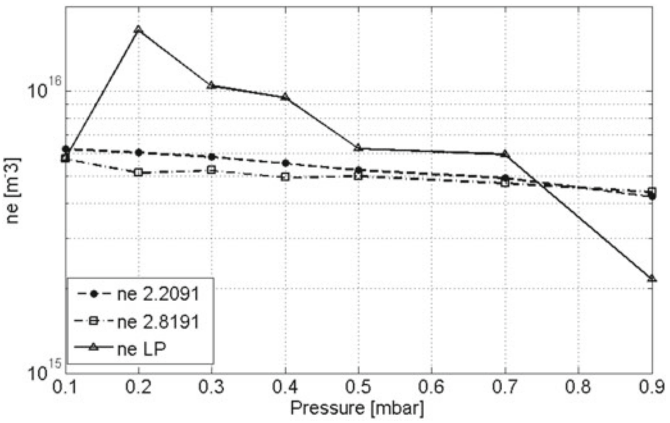



**Fig. 11.** Electron density vs. gas pressure for 70 W of microwave power.

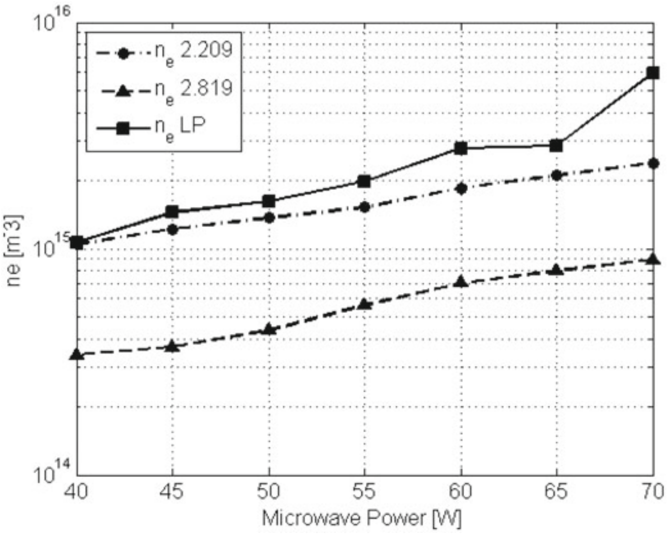



**Fig. 12.** Electron density calculated by means of the frequency shift of the resonant modes for different values of microwave power and at $p = 0.7$ mbar.

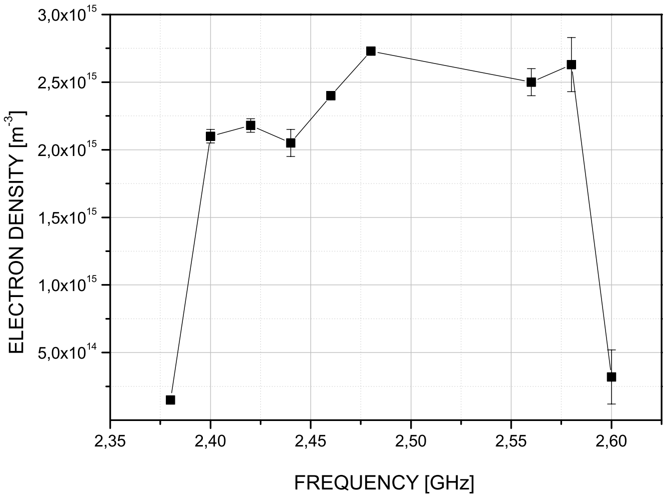



**Fig. 13.** Langmuir probe measured electron density at 10.5 cm inside the cavity for different heating frequencies: $p = 1.1$ mbar, PRF = 70 W.

the equations (1)–(4) we can calculate the electron density from the frequency shift, and compare the results with the Langmuir probe measurements.

The trend of the electron densities coming from the two modes' frequency shift is in reasonable agreement with the one determined by means of the Langmuir probe (Fig. 11), except for the lowest pressure value, for which the Langmuir probe measurement features a strong fluctuation due to the marked inhomogeneities of the plasma. This condition, in fact, corresponds to the early stage of the upstream overdense plasma creation mentioned above. The origin of the different slopes in observed and calculated densities occur because LP measurements are local, while the calculated ones are an average over the whole plasma volume. Figure 12 is particularly interesting, it shows that the three density trends with respect to the microwave power are very similar. Note that the densities calculated through the shift of the $TE_{211}$ mode are considerably smaller than the ones calculated by means of the other mode and/or with the Langmuir probe. This may be due to the stronger influence of anisotropies on the $TE_{014}$, whose structure may be completely altered by the plasma and by the probe insertion. Nevertheless, the frequency shift method gives only a global estimation of the plasma density, and the data reported in Figure 12 are useful only as mean values. In this sense, all the local fluctuations of the electron density due to different mechanisms of plasma-wave interaction cannot be detected by the frequency shift method.

A further investigation about the frequency variations effects on the plasma dynamics has been performed by using the broadband microwave generator named TWT. Figure 13 shows a strong variation (more or less one order of magnitude) of the electron density occurring for small variations of the incoming wave frequency. Similarly Figure 14 shows the variation of the electron temperature and of the plasma potential. The temperature fluctuations are less marked with respect to the density; at high pressures (more than 1 mbar in this case) the high electron collisionality makes the density more sensitive with respect to the temperature for any variation of frequency.

## 7 Discussion

The maximum detected frequency shift observed in the experiments is below 4.5%, i.e. $\simeq$100 MHz. Calculations performed for an empty chamber demonstrate that the spacing between subsequent modes decreases as the frequency increases: around 2.45 GHz it is about 20 MHz, decreasing down to few hundreds of kHz for cavities operating at 28 GHz [17]. Therefore around 2.45 GHz frequency variations of the order of several MHz are needed in order to significantly affect the plasma dynamics. This is confirmed by the results shown in Figures 13: pronounced fluctuations of plasma density are evident only over large variations

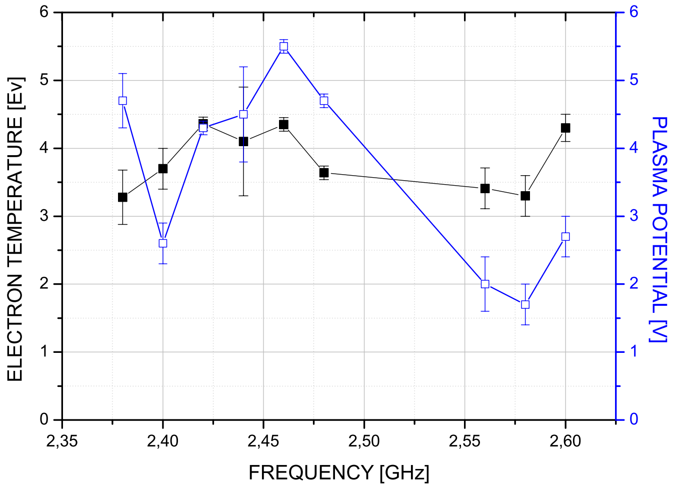



**Fig. 14.** (Color online) Langmuir probe measured electron temperature at 10.5 cm inside the cavity for different heating frequencies: $p = 1.1$ mbar, $PRF = 70$ W.

of the pumping frequency (more than 200 MHz). The frequency tuning, and/or the variation of the cavity length (which leads to similar effects), may improve the wave-plasma energy transfer, thus maximizing the plasma density. Variations of microwave coupling have been observed also when the coaxial connector was used as microwave input: the plasma was ignited with just 5−6 W of rf power. In addition, the density distribution and the plasma temperature changed when the Langmuir probe was moved inside the chamber. These results are interesting even for ECRIS, because in those devices biased disks are usually employed to increase the electron density [18]: the variation of the biased disk position may strongly perturb the plasma dynamics as a consequence of the microwave propagation perturbation, and this effect may be superimposed to the usual effects provided by the biased electrodes. On the other hand, the role of such a disk could be to compensate the frequency shift and optimize some preferential modes. In conclusion, the experiment reported here has demonstrated that standing wave formation occurs inside the cavity, also in the case of high density plasmas and in presence of two ECR zones. The same measurements with other diagnostic devices will be performed in future with the resonant cavities of high performance ECRIS, in order to confirm the results even in case of higher frequencies. Theoretical analysis on the mode formation must be extended to the presence of complex plasma structures: variation of density along the plasma axis and radius, presence of large gradients of magnetic field, etc. have to be taken into account.

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